

Hae-Na Lee

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Research Interests

Human-computer interaction, human-centered AI, accessible computing, computer vision, machine learning

Education

Stony Brook University

PH.D. IN COMPUTER SCIENCE

Stony Brook, NY, USA

Aug 2016 – May 2023

- Advisor: Prof. I.V. Ramakrishnan
- Thesis: *Enhancing the Usability of Computer Applications for People with Visual Impairments via UI Augmentation*

Seoul National University

M.S. IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

Seoul, South Korea

Mar 2013 – Feb 2015

- Advisor: Prof. Suk I. Yoo
- Thesis: *Image Classification using SVM Classifier Learned by AdaBoost Method*

Sookmyung Women's University

B.S. IN COMPUTER SCIENCE

Seoul, South Korea

Mar 2009 – Feb 2013

- *Magna Cum Laude*

Professional Experience

Michigan State University

ASSISTANT PROFESSOR

East Lansing, MI, USA

Aug 2023 – Present

- Department of Computer Science and Engineering

Stony Brook University

GRADUATE RESEARCH ASSISTANT

Stony Brook, NY, USA

Jan 2019 – May 2023

- Knowledge Systems Laboratory

Meta Platforms, Inc.

SOFTWARE ENGINEER INTERN

New York, NY, USA

May 2022 – Aug 2022

- Instagram Relevance Integrity Team

Broadridge Financial Solutions

RESEARCH INTERN

Brentwood, NY, USA

Jun 2019 – Aug 2019

- Broadridge Innovation Laboratory

Korea Institute of Science and Technology

RESEARCH INTERN

Seoul, South Korea

Mar 2015 – Aug 2016

- Center for Robotics Research

Seoul National University

GRADUATE RESEARCH ASSISTANT

Seoul, South Korea

Mar 2013 – Feb 2015

- Artificial Intelligence and Computer Vision Laboratory

Publications

PEER-REVIEWED INTERNATIONAL CONFERENCE

- [C.21] Akshay Kolgar Nayak, Yash Prakash, Sampath Jayarathna, **Hae-Na Lee**, Vikas Ashok. 2026. Contextual Scaffolding and Self-Efficacy: Supporting Computer Skill Development among Blind Learners in India.

Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI). (Accepted) (Acceptance rate: 25.3%)

- [C.20] Yash Prakash, Akshay Kolgar Nayak, Nithiya Venkatraman, Sampath Jayarathna, **Hae-Na Lee**, Vikas Ashok. 2026. [Micro-Behavioral Analysis of Online Shopping Patterns for Blind Users](#). Proceedings of the 2026 ACM SIGIR Conference on Human Information Interaction And Retrieval (CHIIR).
- [C.19] Yash Prakash, Akshay Kolgar Nayak, Shoaib Mohammed Alyaan, Sampath Jayarathna, **Hae-Na Lee**, Vikas Ashok. 2026. [VoxVista: Enhancing Screen Reading Experience for Online User Comments](#). Proceedings of the 2026 ACM SIGIR Conference on Human Information Interaction And Retrieval (CHIIR).
- [C.18] Akshay Kolgar Nayak, Yash Prakash, Md Javedul Ferdous, Sampath Jayarathna, **Hae-Na Lee**, Vikas Ashok. 2026. [Examining Inclusive Computing Education for Blind Students in India](#). Proceedings of the 57th ACM Technical Symposium on Computer Science Education (SIGCSE TS 2026). (Acceptance rate: 30%)
- [C.17] Nithiya Venkatraman, Akshay Kolgar Nayak, Suyog Dahal, Yash Prakash, **Hae-Na Lee**, Vikas Ashok. 2025. [AccessMenu: Enhancing Usability of Online Restaurant Menus for Screen Reader Users](#). Proceedings of the 22nd International Web for All Conference (W4A), pp. 36-47. (**W4A 2025 Best Technical Paper Candidate** 🏆)
- [C.16] Mohan Sunkara, Akshay Kolgar Nayak, Sandeep Kalari, Yash Prakash, Sampath Jayarathna, **Hae-Na Lee**, Vikas Ashok. 2025. [Adapting Online Customer Reviews for Blind Users: A Case Study of Restaurant Reviews](#). Proceedings of the 22nd International Web for All Conference (W4A), pp. 135-146. (**W4A 2025 Best Technical Paper Award** 🏆)
- [C.15] Yash Prakash, Akshay Kolgar Nayak, Shoaib Mohammed Alyaan, Pathan Aseef Khan, **Hae-Na Lee**, Vikas Ashok. 2024. [Improving Usability of Data Charts in Multimodal Documents for Low Vision Users](#). Proceedings of the 26th International Conference on Multimodal Interaction (ICMI), pp. 498-507.
- [C.14] Yash Prakash, Akshay Kolgar Nayak, Sampath Jayarathna, **Hae-Na Lee**, Vikas Ashok. 2024. [Understanding Low Vision Graphical Perception of Bar Charts](#). The 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), Article 59, pp. 1-10. (Acceptance rate: 30%)
- [C.13] Mohan Sunkara, Akshay Kolgar Nayak, Sandeep Kalari, Satwik Ram Kodandaram, Sampath Jayarathna, **Hae-Na Lee**, Vikas Ashok. 2024. [Assessing the Accessibility and Usability of Web Archives for Blind Users](#). The 28th International Conference on Theory and Practice of Digital Libraries (TPDL), pp. 203-221.
- [C.12] Tien Tran, **Hae-Na Lee**, Ji Hwan Park. 2024. [Discovering Accessible Data Visualizations for People with ADHD](#). Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI), Article 64, pp. 1-19. (Acceptance rate: 26.4%, **CHI 2024 Best Paper Honorable Mention Award (Top 5%)** 🏆)
- [C.11] Yash Prakash, Mohan Sunkara, **Hae-Na Lee**, Sampath Jayarathna, Vikas Ashok. 2023. [AutoDesc: Facilitating Convenient Perusal of Web Data Items for Blind Users](#). Proceedings of the 28th International Conference on Intelligent User Interfaces (IUI), pp. 32-45. (Acceptance rate: 24.1%)
- [C.10] **Hae-Na Lee**, Yash Prakash, Mohan Sunkara, I.V. Ramakrishnan, Vikas Ashok. 2022. [Enabling Convenient Online Collaborative Writing for Low Vision Screen Magnifier Users](#). Proceedings of the 33rd ACM Conference on Hypertext and Social Media (HT), pp. 143-153. (Acceptance rate: 36.0%)
- [C.9] **Hae-Na Lee**, Vikas Ashok. 2022. [Impact of Out-of-Vocabulary Words on the Twitter Experience of Blind Users](#). Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI), Article 608, pp. 1-20. (Acceptance rate: 26.0%)
- [C.8] Javedul Ferdous, **Hae-Na Lee**, Sampath Jayarathna, Vikas Ashok. 2022. [InSupport: Proxy Interface for Enabling Efficient Non-Visual Interaction with Web Data Records](#). Proceedings of the 27th International Conference on Intelligent User Interfaces (IUI), pp. 49-62. (Acceptance rate: 24.5%)
- [C.7] **Hae-Na Lee**, Vikas Ashok. 2021. [Towards Enhancing Blind Users' Interaction Experience with Online Videos via Motion Gestures](#). Proceedings of the 32nd ACM Conference on Hypertext and Social Media (HT), pp. 231-236.
- [C.6] **Hae-Na Lee**, Sami Uddin, Vikas Ashok. 2020. [TableView: Enabling Efficient Access to Web Data Records for Screen-Magnifier Users](#). The 22nd International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), Article 23, pp. 1-12. (Acceptance rate: 27.5%)

- [C.5] **Hae-Na Lee**, Vikas Ashok, I.V. Ramakrishnan. 2020. [Repurposing Visual Input Modalities for Blind Users: A Case Study of Word Processors](#). IEEE International Conference on Systems, Man, and Cybernetics (SMC), pp. 2714-2721.
- [C.4] **Hae-Na Lee**, Sami Uddin, Vikas Ashok. 2020. [iTOC: Enabling Efficient Non-Visual Interaction with Long Web Documents](#). IEEE International Conference on Systems, Man, and Cybernetics (SMC), pp. 3799-3806.
- [C.3] **Hae-Na Lee**, Vikas Ashok, I.V. Ramakrishnan. 2020. [Rotate-and-Press: A Non-visual Alternative to Point-and-Click?](#). International Conference on Human-Computer Interaction (HCI International), pp. 291-305.
- [C.2] **Hae-Na Lee**, Vikas Ashok. 2020. [Towards Personalized Annotation of Webpages for Efficient Screen-Reader Interaction](#). Proceedings of the 31st ACM Conference on Hypertext and Social Media (HT), pp. 111-116.
- [C.1] Deokyoung Kang, **Hae-Na Lee**, Suk I. Yoo. 2014. [Detecting Defects in Repeatedly Patterned Image with Spatially Different Level of Noise](#). IEEE International Conference on Image Processing (ICIP), pp. 3258-3262.

PEER-REVIEWED INTERNATIONAL JOURNAL

- [J.10] Nithiya Venkatraman, Anand Ravi Aiyer, Yash Prakash, Sampath Jayarathna, **Hae-Na Lee**, Vikas Ashok. 2025. [Characterizing Language Use in Online Accessibility Discussion Forums](#). ACM Transactions on the Web (TWB).
- [J.9] Javedul Ferdous, Akshay Kolgar Nayak, Yash Prakash, Nithiya Venkatraman, Sampath Jayarathna, **Hae-Na Lee**, Vikas Ashok. 2025. [Understanding Online Discussion Experiences of Blind Screen Reader Users](#). International Journal of Human-Computer Interaction (IJHCI).
- [J.8] Akshay Kolgar Nayak, Yash Prakash, Sampath Jayarathna, **Hae-Na Lee**, Vikas Ashok. 2025. [Insights in Adaptation: Examining Self-reflection Strategies of Job Seekers with Visual Impairments in India](#). Proceedings of the ACM on Human-Computer Interaction (CSCW), Article CSCW304, pp. 1-30.
- [J.7] Brooke Wolfe, Yoo Jung Oh, Hyesun Choung, Xiaoran Cui, Joshua Weinzapfel, R. Amanda Cooper, **Hae-Na Lee**, Rebecca Lehto. 2025. [Caregiving Artificial Intelligence Chatbot for Older Adults and Their Preferences, Well-Being, and Social Connectivity: Mixed-Method Study](#). Journal of Medical Internet Research (JMIR), Vol. 27:e65776.
- [J.6] Yash Prakash, Pathan Aseeef Khan, Akshay Kolgar Nayak, Sampath Jayarathna, **Hae-Na Lee**, Vikas Ashok. 2025. [Towards Enhancing Low Vision Usability of Data Charts on Smartphones](#). IEEE Transactions on Visualization and Computer Graphics (TVCG), Vol. 31, pp. 853-863.
- [J.5] Yash Prakash, Akshay Kolgar Nayak, Mohan Sunkara, Sampath Jayarathna, **Hae-Na Lee**, Vikas Ashok. 2024. [All in One Place: Ensuring Usable Access to Online Shopping Items for Blind Users](#). Proceedings of the ACM on Human-Computer Interaction (EICS), Article 257, pp. 1-25.
- [J.4] Mohan Sunkara, Yash Prakash, **Hae-Na Lee**, Sampath Jayarathna, Vikas Ashok. 2023. [Enabling Customization of Discussion Forums for Blind Users](#). Proceedings of the ACM on Human-Computer Interaction (EICS), Article 176, pp. 1-20.
- [J.3] Javedul Ferdous, **Hae-Na Lee**, Sampath Jayarathna, Vikas Ashok. 2023. [Enabling Efficient Web Data-Record Interaction for People with Visual Impairments via Proxy Interfaces](#). ACM Transactions on Interactive Intelligent Systems (TiiS), Article 13, pp. 1-27.
- [J.2] **Hae-Na Lee**, Vikas Ashok. 2022. [Customizable Tabular Access to Web Data Records for Convenient Low-Vision Screen Magnifier Interaction](#). ACM Transactions on Accessible Computing (TACCESS), Article 14, pp. 1-22.
- [J.1] **Hae-Na Lee**, Vikas Ashok, I.V. Ramakrishnan. 2021. [Bringing Things Closer: Enhancing Low-Vision Interaction Experience with Office Productivity Applications](#). Proceedings of the ACM on Human-Computer Interaction (EICS), Article 197, pp. 1-18.

LIGHTLY PEER-REVIEWED PUBLICATIONS (POSTERS, EXTENDED ABSTRACTS, WORKSHOP PAPERS, AND TECHNICAL REPORTS)

- [P.2] Mohan Sunkara, Akshay Kolgar Nayak, Sandeep Kalari, Yash Prakash, Sampath Jayarathna, **Hae-Na Lee**, Vikas Ashok. 2025. [QuickQue: Enabling Quick Access to Information in User Reviews for Screen Reader Users](#). Proceedings of the 22nd International Web for All Conference (W4A), pp. 22-24. Technical Note.
- [P.1] **Hae-Na Lee**, Vikas Ashok, I.V. Ramakrishnan. 2020. [Screen Magnification for Office Applications](#). The 22nd International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), Article 95, pp. 1-3. Demonstration.

Honors and Awards

- 2025 **Best Technical Paper Candidate**, ACM 22nd International Web for All (W4A) Conference [C.17]
Best Technical Paper Award, ACM 22nd International Web for All (W4A) Conference [C.16]
- 2024 **Best Paper Honorable Mention Award**, ACM Conference on Human Factors in Computing Systems (CHI) [C.12]
- 2019 **Excellence as a TA Award**, Department of Computer Science, Stony Brook University
- 2016 – 2019 **University Fellowship**, Stony Brook University
- 2013 – 2015 **Global PhD Fellowship**, National Research Foundation of Korea
 Research Topic: *Image Classification using Sparse and Hierarchical Representation of Classified Object Image Data*
- 2013 **Academic Excellence Scholarship**, Seoul National University
- 2009 – 2012 **Academic Excellence Scholarship**, Sookmyung Women's University

Grants

- 2024 **Trifecta Initiative Facilitating Funds (TIFF) Award**, Michigan State University. Older Adults' Preferences for Caregiving AI Chatbots to Improve Well-being and Social Connectivity. 01/2024–12/2024. Total: \$9,708. PI: Yoo Jung Oh, Co-PIs: Hyesun Choung, Brook Wolfe, Rebecca Lehto, **Hae-Na Lee**.

Teaching Experience

Michigan State University

INSTRUCTOR

East Lansing, MI, USA

Jan 2024 – Present

- Fall 2024 – 2025 **CSE 841: Artificial Intelligence**
- Spring 2024 – 2026 **CSE 491: Selected Topics in Computer Science (Human-Computer Interaction)**

Stony Brook University

GRADUATE TEACHING ASSISTANT

Stony Brook, NY, USA

Aug 2016 – Dec 2018

- Fall 2018 **CSE 537: Artificial Intelligence**
- Spring 2018 **CSE 303: Introduction to Theory of Computation**
- Fall 2017 **CSE 215: Foundations of Computer Science I**
- Spring 2017 **CSE 101: Introduction to Computers**
- Fall 2016 **CSE 114: Computer Science I**

Mentoring Experience

ADVISEES

- Ph.D. **Dipannita Biswas (MSU CSE)**, Fall 2025 – Present
- Undergrad **Patrick Lo (MSU CSE)**, Fall 2024 – Present
- Peter Pena (MSU CSE)**, Fall 2024 – Present
- Chirag Bhansali (MSU CSE)**, Fall 2025 – Present

DISSERTATION COMMITTEES

- 2025 **Yash Prakash (Old Dominion University, Ph.D.)**, Thesis: *Enhancing Data Usability for People with Visual Impairments* (defended on July 2025).
- Mohan Sunkara (Old Dominion University, Ph.D.)**, Thesis: *Enhancing Non-visual Interaction with Online User-generated Content* (defended on July 2025).

Professional Service

REVIEWER

- ACM CHI Conference on Human Factors in Computing Systems (CHI)
- ACM International Conference on Intelligent User Interfaces (IUI)
- Springer Nature Scientific Reports
- ACM Transactions on Accessible Computing (TACCESS)
- ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)
- ACM Designing Interactive Systems (DIS)
- ACM Conversational User Interfaces (CUI)
- International Conference on Pattern Recognition and Artificial Intelligence (ICPRAI)

INSTITUTIONAL SERVICE

- MSU CSE Graduate Student Recruiting Committee (GSRC) 2024 – Present
- MSU Designathon Judge 2025